HO JANG

Curriculum Vitae

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EDUCATION & EMPLOYMENT HISTORY

University of Virginia *Aug. 2023 – Present*
Ph.D. in Physics
GPA: 3.973 / 4.0

Gwangju Institute of Science and Technology (GIST) *Mar. 2017 – Aug. 2023*
B.S. in Physics

University of California, Berkeley *Jun. 2018 – Aug. 2018*
Exchange Student

Teaching Assistant, University of Virginia *Aug. 2023 – May 2024*
General Physics Lab I & II
Guided and supported students during lab instruction.

Research Assistant, University of Virginia *May 2024 – Present*
Conducting research in theoretical condensed matter & nuclear physics.

PUBLICATIONS

Reviewed Articles

- Jinheung Kim, Taegyu Lee, Jeongwoo Kim, **Ho Jang**. *Implementation of the ATLAS-SUSY-2018-06 analysis in the MadAnalysis 5 framework.* Modern Physics Letters A (2020). Link
- Diego Ibarra-Hoyos, Peter F. Connors, **Ho Jang**, et al. *Quantum-annealed machine learning discovers ductile, high strength and corrosion-resistant high-entropy alloy. npj Computational Materials* (2026). Link
- Lingyu Yang, **Ho Jang**, Sankha Subhra Bakshi, Yang Yang, Gia-Wei Chern. *Pseudospin Formulation of Quench Dynamics in the Semiclassical Holstein Model.* Physics Review B (2026). Link
- **Ho Jang**, Yang Yang, Gia-Wei Chern. *Anomalous coarsening and nonlinear diffusion of kinks in a one-dimensional quasiclassical Holstein model.* Physical Review E **114**, 014225 (2026). Link.

Preprints / Manuscripts

- **Ho Jang**, Yang Yang, Gia-Wei Chern. *Kinetics of Peierls transition: Machine Learning Force-Field approach* (Oct. 2025). arxiv.org/abs/2510.20659
- **Ho Jang**, Jackson C. Glass, Gia-Wei Chern, *Learning Degenerate Manifolds of Frustrated Magnets with Boltzmann Machines* (Nov. 2025). arxiv.org/abs/2511.19879
- **Ho Jang**, Gia-Wei Chern, *Transformer Learning of Chaotic Collective Dynamics in Many-Body Systems* (Jan. 2026). arxiv.org/abs/2601.19080
- **Ho Jang**, Gia-Wei Chern, *Suppressed coarsening after an interaction quench in the Holstein chain* (Feb. 2026).arxiv.org/abs/2602.05815
- Jitao Xu, **Ho Jang**, Zaki Panjsheeri, et al. *Neural Network Representation of Generalized Parton Distributions (NNGPD).* arXiv:2605.06994 (2026).
- Arunangshu Bora, Sankha Subhra Bakshi, **Ho Jang**, and Gia-Wei Chern. *Cooperative Domain-Wall Dynamics in a Two-Dimensional Quasiclassical Holstein Model.* arXiv:2609.28436 (2026).
- **Ho Jang**, Sankha Subhra Bakshi, and Gia-Wei Chern. *Exact Equivariance from Ordinary Neural Networks for Lattice Many-Body Dynamics.* arXiv:2609.27132 (2026).
- Yang Yang, Ali Rayat, **Ho Jang**, Yunhao Fan, Sankha Subhra Bakshi, and Gia-Wei Chern. *Adiabatic Variational Principle: Born-Oppenheimer Theory for Emergent Electronic-Order Dynamics.* arXiv:2610.40117 (2026).

CONFERENCES & PRESENTATIONS

- **Parton Distribution Function and Lattice Calculation Workshop** (Jefferson Laboratory, VA), Nov. 2024. Poster: *Neural network for Compton Form Factor to Mellin moments without Generalized Parton Distributions.*
- **APS March/April Meeting** (Anaheim, CA), Mar. 2025. Oral: *Anomalous coarsening in one-dimensional quasi-classical Holstein model.*

- **APS Southeastern Section Meeting** (James Madison University, VA), Oct. 2025. Session Chair: *Two-dimensional materials.*
- **APS Southeastern Section Meeting** (James Madison University, VA), Oct. 2025. Oral: *Anomalous coarsening in one-dimensional quasi-classical Holstein model; Kinetics of Peierls transition.*
- **UVa Physics Annual Graduate Student Research Poster Exhibition** (UVa, VA), Mar. 2026. Poster: *Suppressed coarsening in a Holstein chain*
- **APS March/April Meeting** (Denver, CO), Mar. 2026. Oral & Poster: *Suppressed coarsening after an interaction quench in the Holstein chain; Kinetics of Peierls transition: ML-force field approach.*
- **UVa Research Computing Exhibition** (UVa, VA), Apr. 2026. Poster: *Defect dynamics and chaotic trajectory in Many-Body Physics*
- **Towards improved hadron femtography with hard exclusive reactions** (UVa, VA), Jul. 2027. *Attending session with collaborators about NNGPD*
- **MGI PI Meeting** (Washington, DC), Aug. 2027. *Participating in project discussions with collaborators on NNGPD.*

HONORS

- National Scholarship for Science and Engineering (Korea Student Aid Foundation), *Mar. 2017 – Dec. 2022.*
- Academic Excellence Scholarship (GIST), *Mar. 2019 – Dec. 2019.*

TEACHING & MENTORING

- Teaching Assistant, General Physics Lab I & II (University of Virginia), *Aug. 2023 – May 2024.*
- Mentoring Program, Mentee: Bora Arunangshu (Physics Ph.D) *Sep. 2025 – Dec. 2025.*
- Uplift Physicists Program by the National Society of Black Physicists (NSBP), Mentee: Woodson Chance A (Physics undergraduate) *Sep. 2026 – Sep. 2027.*
- DMRG Lecture Series (University of Virginia), *Sep. 2026.* Organized a month-long series of lectures by Nishan Ranabhat on DMRG methods.

SOFTWARE SKILLS

- **Julia**: Large-scale simulations; GPU acceleration (CUDA).
- **Python**: Machine learning (PyTorch, TensorFlow); quantum machine learning (D-Wave).
- **Mathematica / Symbolic computation**: Analytical derivations and prototyping.
- **Scientific software**: Quantum ESPRESSO (density functional theory); ITensor and Cytnx (tensor network calculations); custom Julia and Python code for DMRG,

COLLABORATIONS

- 3dHadron external collaboration, *Aug. 2025 – Present.* Collaborator: Liuti Simonetta. Focus: ML modeling; analytic analysis for CFF reconstruction.
- Experimental condensed matter group, *Aug. 2025 – Present.* Collaborator: Joseph Poon. Focus: quantum machine learning modeling.